

One Pager

PowerPulse

Team Token Titans · Track 04 — Energy & Climate Infrastructure · Frontiers GenAI Hackathon, ALX Kigali

An outage assistant for African households and small businesses. Ask in your own language whether your power cut is planned, how long it's likely to last, and what to do about it for the next few hours — grounded in the utility's own published records, and built to say "I don't know" instead of inventing a restoration time. Rwanda is the MVP.

The problem

When the power goes out, the person sitting in the dark has no idea whether it's planned maintenance ending in two hours or a fault lasting until tomorrow. The utility often publishes the answer — as an HTML table, in English, paginated across 126 pages, on a website that assumes a laptop and a good connection.

The information exists. It does not reach the person who needs it, in the language they speak, on the phone they own. That gap is not a Rwandan problem — it is the default across the continent.

The decisions waiting on that answer are real and time-sensitive: **do I discard the stock in my fridge? do I buy fuel for the generator? do I relocate to finish this work?**

What it does

Ask in your own language — *"nta muriro mu Kimironko"*, *"power out Gisozi"* — and get back three things:

1. **Is it planned?** Matched against the utility's published schedule for your area.
2. **How long?** The published window if one exists. If not, the historical average for *your* area, labelled as historical, with the sample size it rests on.
3. **What to do now** — advice that changes with the outage duration **and** with who you are: household, shop owner, or remote worker.

The third part is the product. Anyone can look up a schedule. The same location at the same moment produces materially different guidance for a shop owner with cold stock than for a remote worker with a laptop battery.

Why generative AI is load-bearing

Not a chatbot bolted onto a table. Three places where nothing else works:

1. **Messy, code-switched, multilingual input.** *"nta muriro mu Kimironko"*, *"power out Gisozi"*, *"hano mu Masaka nta amashanyarazi"* — matched against free-text place names. This is what makes one product work across languages instead of one product per language.
2. **Unstructured source data.** The `Sector / Areas` and `Reason` fields are prose: multiple districts per row, `All sectors in Rutsiro`, feeder names buried in quotes. Every utility publishes its own idiosyncratic mess — a parser generalizes badly, a model generalizes well.

3. **Synthesis, not retrieval.** Duration + profile + food-safety thresholds + historical confidence, composed into advice in the user's language.

Data source — real, public, verifiable

<https://www.reg.rw/customer-service/power-outages/> — Rwanda Energy Group's official outage listing.

502 rows scraped across **126 pages**, exploded into **3,201 records** covering **1,047 (district, sector) pairs**, spanning **24 April 2018 – 5 August 2026** (498 historical, 4 forward-scheduled). One record per outage × affected area, with zero rows dropped in parsing and 55 (11%) flagged as low-confidence rather than silently guessed. Committed to the repo. **A judge can open the source and check us.**

No invented data. No seeded demo records. No cold start — the system is useful the moment it launches.

Why Rwanda first, and why it doesn't stop there

Rwanda is the MVP market, not the scope. REG publishes its outage records openly, which makes it the fastest place to prove the pipeline against real data — and a system that can't survive contact with one real utility's messy HTML won't survive twenty.

The architecture is utility-agnostic by construction: the ingest layer normalizes any published outage listing into the same schema, and the language layer is a prompt configuration rather than a rewrite. **A new country is a new scraper and a new language, not a new product.**

Evaluation results

33 questions: 25 answerable from REG data, 3 where REG published a start time and never an end, and 5 deliberately out of scope.

METRIC	RESULT
Location match accuracy	28 / 28
Correct refusal rate (out-of-scope)	5 / 5
Duration-grounded advice	3 / 3
Honest when no ETA was published	3 / 3
Answers with no invented clock time	28 / 28
Answers carrying a visible source line	28 / 28

Scored deterministically — no model grades its own work. The last two metrics pull against each other on purpose: one punishes ignoring the window the utility published, the other punishes inventing a window it never published. Every `HH:MM` in an answer must also appear in the context the model was given.

Full question set and scoring script in the repo under `eval/`.

Ethics choices

All three bonus criteria, structural rather than bolted on:

- **Human approval step** — when location match confidence is low, the system asks the user to confirm which area they mean and offers the closest candidates. It does not guess. A confidently wrong location is our worst failure mode, so we designed it out.

- **Evaluation metrics** — above. Including a refusal rate, because knowing when to decline is a feature we scored.
- **Personalization with constraints** — advice varies by profile and duration, under explicit written constraints published in the repo as `prompts/system.txt` : never state a restoration time the utility has not published; never invent facts about the world; label estimates below $n=5$ as weak; bias conservative on food, stock and money.

Food-safety thresholds live in `data/food_safety.json` — the model explains them, it never calculates them. When a call is marginal we recommend discarding. **We accept waste over illness, and we say so.**

We built it to say “I don’t know.” Most demos would hallucinate an ETA to look smooth. Ours shows a real no-ETA case on purpose.

Track fit

Track 04 asks for a tool that helps a household adapt to energy conditions. That is the product, stated literally — a household or small business deciding what to do about a power cut, in the hours it is happening.

It satisfies **Track 03** on every count as well: **speaks the user’s language, 390px mobile viewport, designed as an SMS/USSD surface** for a low-end device on an unreliable connection. All three of that track’s user constraints, not just one. We built for those constraints because our user has them — not to qualify for a track.

What’s next

- **The next utility** — the ingest layer is built to normalize a second published outage listing without touching the reasoning layer. Proving that is the first post-hackathon milestone.
- **SMS/USSD in production** — reaches a feature phone with no data connection, which is the actual constraint for most of the continent.
- **Unplanned outages** — utilities post these to social media without ETAs; ingesting them is where honest uncertainty matters most.
- **More languages** — the pipeline is language-agnostic; Swahili, French and Amharic are the same work repeated, not new engineering.
- **Area-level reliability profiles** — with enough history, “your area averages 7 outages a year” becomes a planning tool, not just an alert.